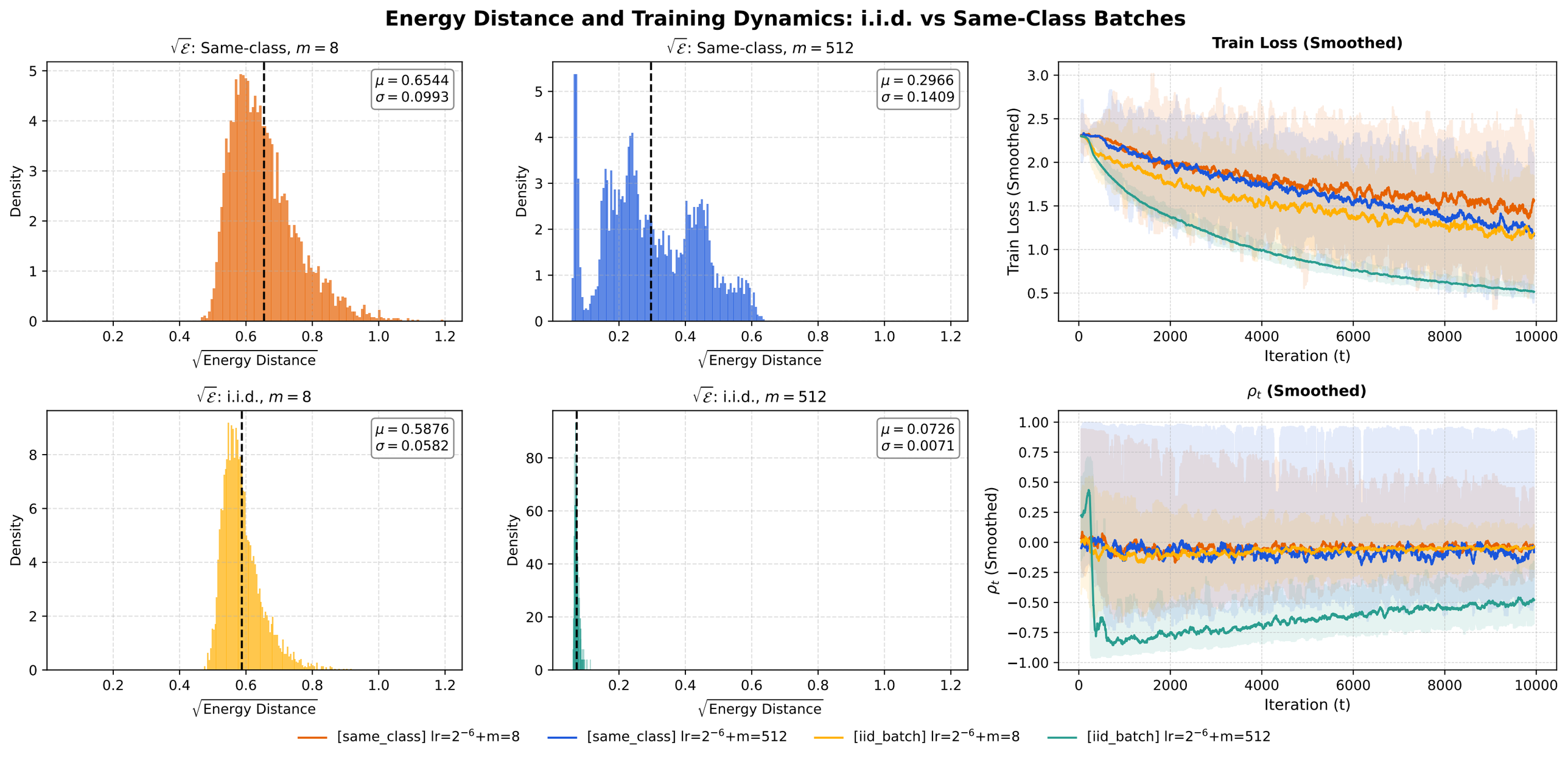

Energy Distance and Training Dynamics: i.i.d. vs Same-Class Batches
$\sqrt{\mathcal{E}}$: Same-class, $m = 8$
$\mu = 0.6544$
$\sigma = 0.0993$
$\sqrt{\mathcal{E}}$: Same-class, $m = 512$
$\mu = 0.2966$
$\sigma = 0.1409$
Train Loss (Smoothed)
Train Loss (Smoothed)
Iteration (t)
$\sqrt{\mathcal{E}}$: i.i.d., $m = 8$
$\mu = 0.5876$
$\sigma = 0.0582$
$\sqrt{\mathcal{E}}$: i.i.d., $m = 512$
$\mu = 0.0726$
$\sigma = 0.0071$
ρ_t (Smoothed)
ρ_t (Smoothed)
Iteration (t)
Density
$\sqrt{\text{Energy Distance}}$
[same_class] lr=2^{-6}+m=8
[same_class] lr=2^{-6}+m=512
[iid_batch] lr=2^{-6}+m=8
[iid_batch] lr=2^{-6}+m=512